

1.(2023 AMC8 Problem 22) In a sequence of positive integers, each term after the second is the product of the previous two terms. The sixth term in the sequence is 4000. What is the fourth term?

2.N is a four-digit number. We are given that all of N's digits are prime, $N \bmod 11$ is 0, and that N is divisible by 36. Find the number of divisors of N.

翻译：

1. (2023 AMC8 问题 22) 在正整数序列中，第二个之后的每个项都是前两个项的乘积。序列中的第六项是 4000。第四个任期是什么？
2. N 是一个四位数字。我们已知 N 的所有数字都是质数， $N \bmod 11$ 为 0，且 N 可被 36 整除。 N 有多少个因数？

Yijian Math Circle